

Aditya Singh

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SUMMARY

Backend Engineer at TCS with 2 year owning Python/FastAPI microservices in production – processing hundreds of thousands of records daily across 3+ business units, 99%+ uptime, 40% query latency reduction. CS grad with an AI major; strong in SQL optimization at scale, data pipeline design, CI/CD automation, LLM integration and RAG.

EXPERIENCE

Backend Engineer

Tata Consultancy Services

Sep. 2024 – Present

Noida, UP

- Architected and owned 5+ Python/SQL microservices processing high-volume enterprise workloads across 3+ business units in an Agile environment, consistently meeting sprint deadlines and SLA targets.
- Optimized 100+ SQL queries through indexing and restructuring, cutting data retrieval latency by ~40% on prod db.
- Designed and maintained 10+ REST APIs with FastAPI – enforcing request validation, structured error handling, and versioning, reducing downstream integration failures by ~30%.
- Wrote unit and integration tests with pytest, pushing coverage from 0% to 70%+ on two core modules; caught and resolved critical regressions before every production release.
- Shipped 2 production modules with 3 cross-functional teams on schedule, serving 500+ end users; maintained 99%+ uptime resolving 100+ incidents within SLA.
- Containerized two services with Docker and built GitHub Actions CI pipelines automating pytest suites and Docker builds on every PR – reducing engineer onboarding from 8 hours to under 2 and eliminating manual pre-deploy errors.

EDUCATION

Bennett University

B.Tech in Computer Science Engineering – Major in Artificial Intelligence | CGPA: 8.32/10

Greater Noida, UP

Aug. 2020 – Jun. 2024

- Relevant Coursework: Machine Learning, Deep Learning, DBMS, Data Structures & Algorithms, Operating Systems, OOP, Computer Networks, System Design

PROJECTS

Power Vocab App | React, TypeScript, Firebase, OpenAI API, Tailwind CSS, Framer Motion

2026

- Built and deployed a full-stack vocabulary learning web app on Firebase Hosting with Google OAuth + email auth, protected routes, and real-time Firestore sync.
- Integrated OpenAI API for contextual AI sentence generation per word; built an animated card-based Learn mode with per-word progress tracking and a save-for-revision flow.
- Engineered live user stats (streak, words learned, quiz scores) via React Context and Firestore, surfaced on a Dashboard with Word of the Day and an adaptive MCQ quiz module.

AI Data Analyst Copilot | Python, FastAPI, OpenRouter API, LangChain, FAISS, RAG, Streamlit

2026

- Built and deployed an AI-powered data analysis tool on Streamlit Cloud where users query CSV datasets in plain English – LLM converts input to executable Pandas code and auto-generates Matplotlib visualizations.
- Engineered a code sanitization and validation pipeline to clean LLM-generated code before execution, reducing runtime errors by ~65% and improving response reliability across diverse dataset structures.
- Built a RAG pipeline using FAISS vector search over dataset schema and sample rows – grounding LLM prompts with live context via LangChain, cutting hallucinated column references by ~60% and enabling multi-turn analyst sessions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Flask, React, Streamlit, LangChain, TailwindCSS

Databases: Supabase, PostgreSQL, MySQL, Firebase Firestore, SSMS, Redis – indexing, query optimization

Tools & Platforms: Git, Docker, Postman, Linux/Bash, Pytest, JIRA, Agile/Scrum, Brevo, Vercel, Render

CI/CD & Cloud: GitHub Actions – test pipelines, Docker builds, PR checks | Firebase Hosting, Streamlit Cloud

AI / ML: LLM APIs, RAG (FAISS), LangChain, Prompt Engineering, Pandas, NumPy, Scikit-learn, TensorFlow

CERTIFICATIONS

Deep Learning Specialization – DeepLearning.AI (2022) · Mastering System Design – Udemy (2026) · Data Structures and Algorithms – Udemy (2026) · Introduction to Computer Vision – IBM (2022) · Azure AI Essentials – Microsoft (2026)